

EXTRACTMET

SmartMelt

A hybrid physics + machine-learning mathematical model for
real-time melt control in Indian MSME steel plants

MATHEMATICAL MODEL DOCUMENTATION · PYTHON REFERENCE IMPLEMENTATION · OPERATOR CONSOLE

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WHY A MODEL, NOT A DASHBOARD

Every melt shop already has a control system. None of them have a plant.

50–120 kWh/t

recoverable energy loss per heat — the gap between average practice and the plant's own best heats, same equipment, same scrap

$\pm 15\text{--}25\text{ }^{\circ}\text{C}$, $\pm 0.02\text{--}0.05\text{ \%C}$

typical endpoint scatter on manual practice — driving over-refining, reblows, and reject heats

2,000+ heats

of Industry-Y BOF data behind the validated endpoint model this system's approach is built on

The gap is not measurement — it's a model of the heat that turns sparse, noisy sensor readings into a trustworthy prediction of what the bath is doing right now, and what it will do if nothing changes.

Three layers, one principle: ML corrects physics, it never replaces it

01

Physics core

फिजिक्स कोर

Coupled mass, energy, species and lining ODE. Thermochemical equilibria, kinetics, slag model. Runs in microseconds — the load-bearing structure that never drifts silently.

02

ML residual layer

एमएल अवशिष्ट परत

Learns $y_{\text{measured}} - f_{\text{physics}}$ only. Gaussian-process + gradient-boosted residual, maturity tiers from cold-start (200 heats) to calibrated (2,000+). Bounded extrapolation by construction.

03

Estimation & control

अनुमान और नियंत्रण

Extended Kalman Filter fuses sparse sensors into a live state estimate. Charge-mix optimiser and receding-horizon MPC issue advisory recommendations — no PLC write path.

The physics core: 30 governing equations, one state vector

E1–E5

Balances

Mass (metal + slag species), bath & solid energy, N-node radial lining conduction

E6–E13

Equilibria

logK for [C],[Si],[Mn] oxidation, slag optical basicity, activity coefficients, h_O

E14–E19

Endpoints

%C/%Si/%Mn equilibrium floors, Healy P-partition, sulphide capacity, L_S

E20–E23

Energy

Theoretical melt floor, IF coupling vs. fill, EAF arc efficiency, converter losses

E24–E30

Kinetics

Rate laws for decarburisation, Si/Mn/P/S transfer, oxygen ledger with slag saturation

E50–E60

Diagnostics

Standing-loss UA test, identifiability, energy audit closure, calibration bounds

Every plant-specific number — furnace geometry, lining thickness, coupling efficiency, sensor suite — lives in one YAML config. The equations never change between an IF Lite install and an EAF/BOF Pro install.

$$y = f_{\text{physics}}(x, \theta) + g_{\text{ML}}(\text{features}) + \varepsilon \text{ — and why that order matters}$$

Residual learning (E31–E39)

- Matern-3/2 Gaussian process on the physics residual, clipped to a physically bounded correction
- Quantile gradient-boosted trees (HistGBM) for endpoint intervals as data accumulates
- Maturity tiers: cold-start (<200 heats, physics-led) → deployable (1,000) → calibrated (2,000+)
- Population Stability Index + MAPE drift monitor — flags regime change (new scrap supplier) before it silently degrades advice
- Tramp-element soft sensor (Cu, Sn, Cr): mass-balance prior + Ridge correction

State estimation (E32–E33)

- Extended Kalman Filter over an augmented [state, θ] vector — joint tracking of bath conditions and slow plant parameters
- $\theta = \{\eta_{\text{electrical}}, UA_{\text{lining_scale}}, k_{\text{C_scale}}, \dots\}$ — the same parameters an offline calibration fits, now tracked online
- Finite-difference Jacobians, Joseph-form covariance update for numerical stability
- Sensor masking: a Lite install (immersion TC only) and a Pro install (+ off-gas, sub-lance) share one filter — just a different observation model

From estimate to action: least-cost charge, receding-horizon control

Charge-mix optimiser (E40–E45)

$\min \sum \text{price} \cdot \text{mass}$ s.t. grade chemistry window, hard tramp-element walls, integer lot sizes

- Solved as an LP/MILP (scipy linprog/milp) — plant-agnostic; only the material table and constraints change per client
- Hard walls on Cu/Sn/Cr recovery=1 (no metallurgical sink) — the constraint a spreadsheet mix can't enforce
- Shadow prices reported per tramp-element constraint: the ₹/t value of relaxing it — the number that sells the case for better scrap sorting
- Pays back from heat #1 — no training history required, unlike the ML layer

Melt MPC (E50–E53)

- Receding-horizon SLSQP over piecewise-constant power/O₂/addition blocks (4–6 per horizon)
- Projects onto the furnace's real discrete tap-power levels — no continuous setpoint a controller can't realise
- Refractory and overheat barrier terms keep recommendations inside safe operating envelope
- `tap_time_advice()`: the single number an operator actually wants — output is a recommendation, not a setpoint write
- Deliberately advisory-only in this phase: builds the trust and the logged-outcome dataset a future closed-loop mode would need

VALIDATION DISCIPLINE

$\pm 15\text{ }^{\circ}\text{C}$ and $\pm 0.02\text{ \%C}$ only mean something with three qualifiers attached

1-sigma, 2-sigma, or max?

`endpoint_hit_rate()` reports hit-rate at a stated tolerance plus bias, p95 error — never a bare number

On what split?

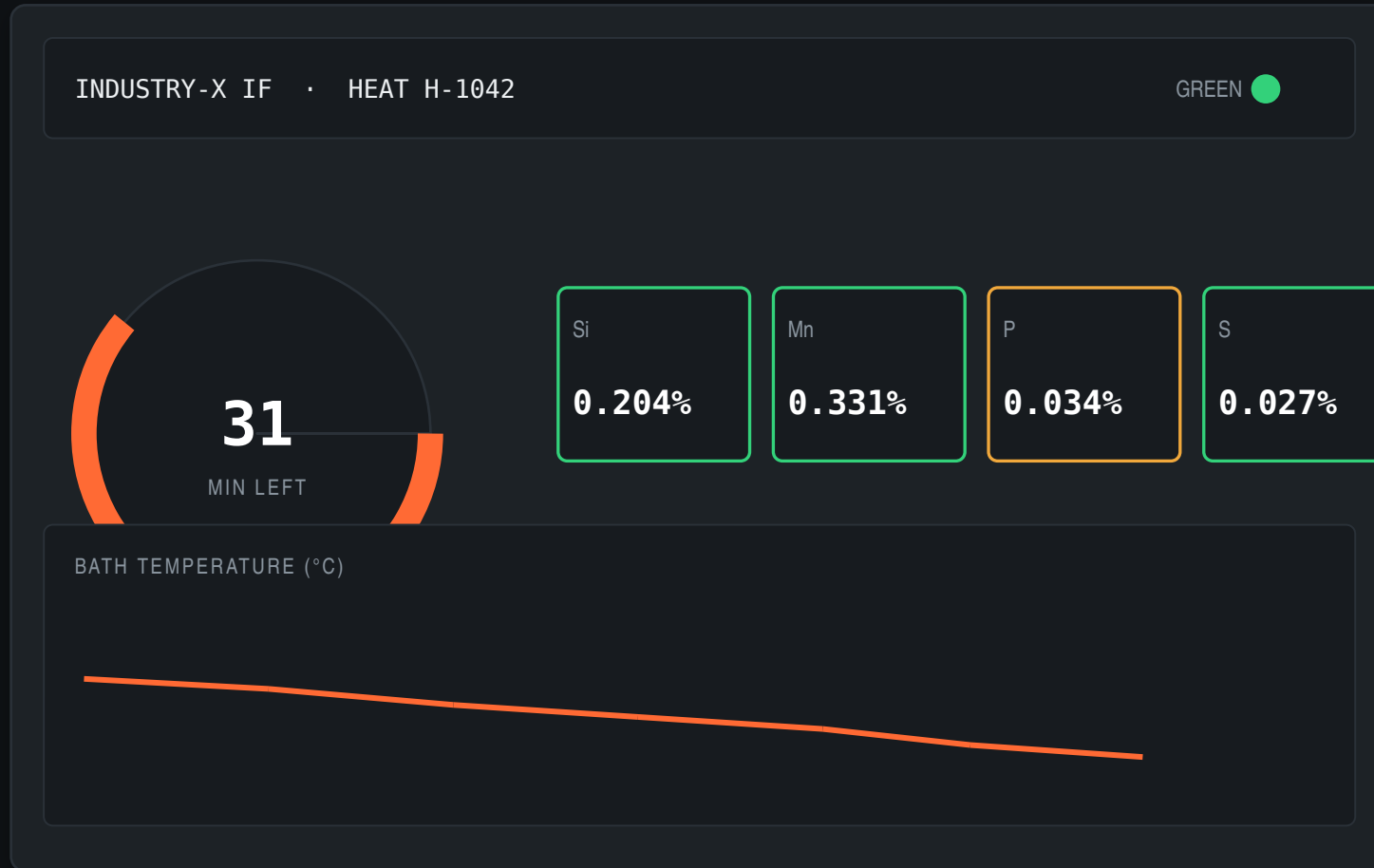
time-ordered train/test only — heats 1..N train, N+1..M test. Random k-fold leaks the lining campaign and the scrap regime

Against what reference?

$\sigma_{\text{model}}^2 = \sigma_{\text{observed}}^2 - \sigma_{\text{reference}}^2$ — a drop-cell has its own $\pm 3\text{--}5^{\circ}\text{C}$, and that's not the model's error

Plus one cheap experiment: a 20–30 min standing-loss hold (power off, full bath) directly identifies `UA_lining_scale` and de-correlates it from $\eta_{\text{electrical}}$ in every later fit — the difference between a fitted model and a curve-fit.

One screen: where the heat is, where it's going, what to do



Heat clock

Radial dial: elapsed / remaining time, live process stage — charging, melting, refining, tap-ready

Live trends

Bath temperature and carbon vs. tap aim $\pm$ tolerance band, updating from the state estimate

Instrument grid

Si / Mn / P / S / slag B2 / FeO — each tile colours green / amber / red against its own tolerance

Energy audit

Where every grid kWh went this heat: useful melt vs. lining, cooling, radiation, converter loss

Advisory feed

Bilingual (EN/Hindi) recommendation log with the reason attached — never a bare instruction

Economics are conservative by construction, and the roadmap is honest

Every saving number is a disputed line, shown

- Achievable target = the plant's OWN best-decile heat, same scrap, same equipment — not a vendor benchmark
- `savings_attribution()`: regression-adjusts for charge mass / carbon / tap-T drift between the before/after windows before crediting SmartMelt
- Payback shown with tariff, opex, and carbon-credit lines broken out — $\text{capex} \div \text{net annual}$, in months
- 5-year NPV at 12% discount reported alongside payback, not instead of it

Roadmap

Now

Advisory-only Phase 1: physics core + charge-mix optimiser ship from heat #1, no history needed

200–1,000 heats

ML residual layer moves cold-start → deployable; drift monitor active from day one

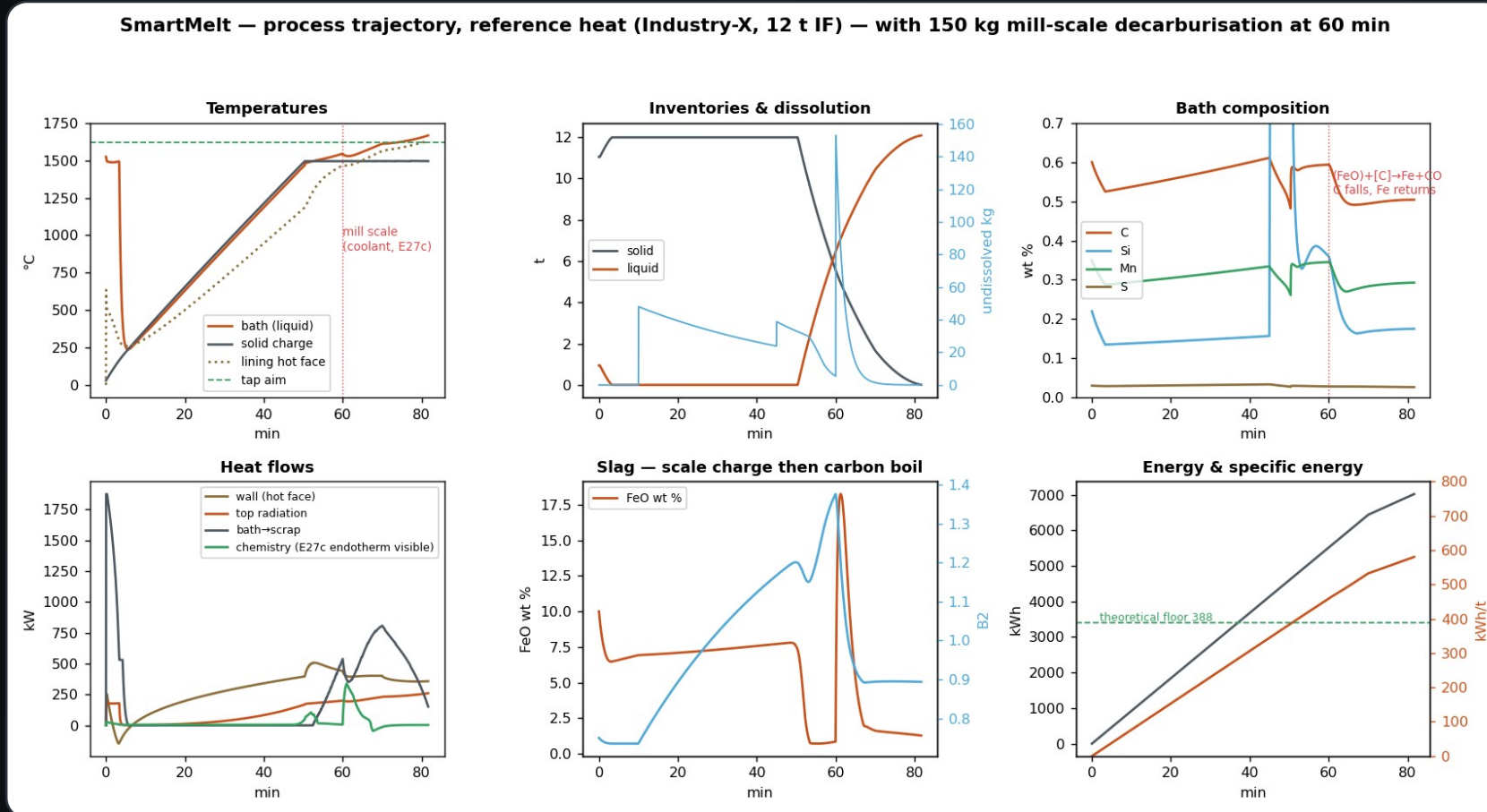
2,000+ heats

Calibrated tier; MPC recommendations tighten; shared-savings evidence log mature enough to bill against

Future

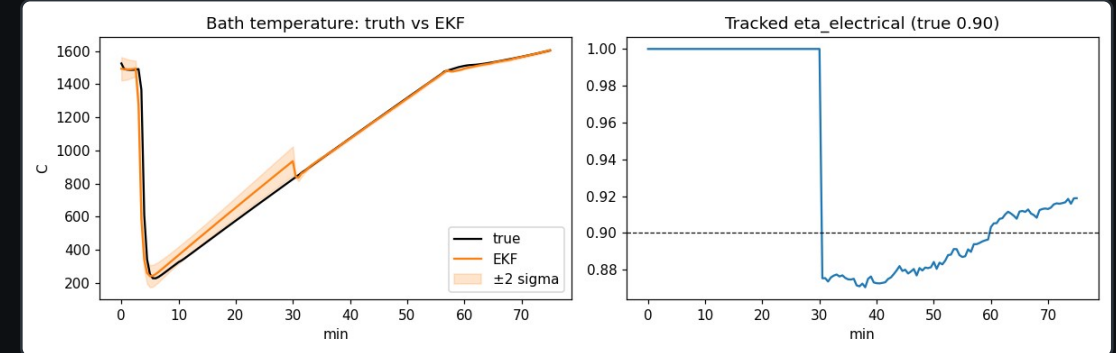
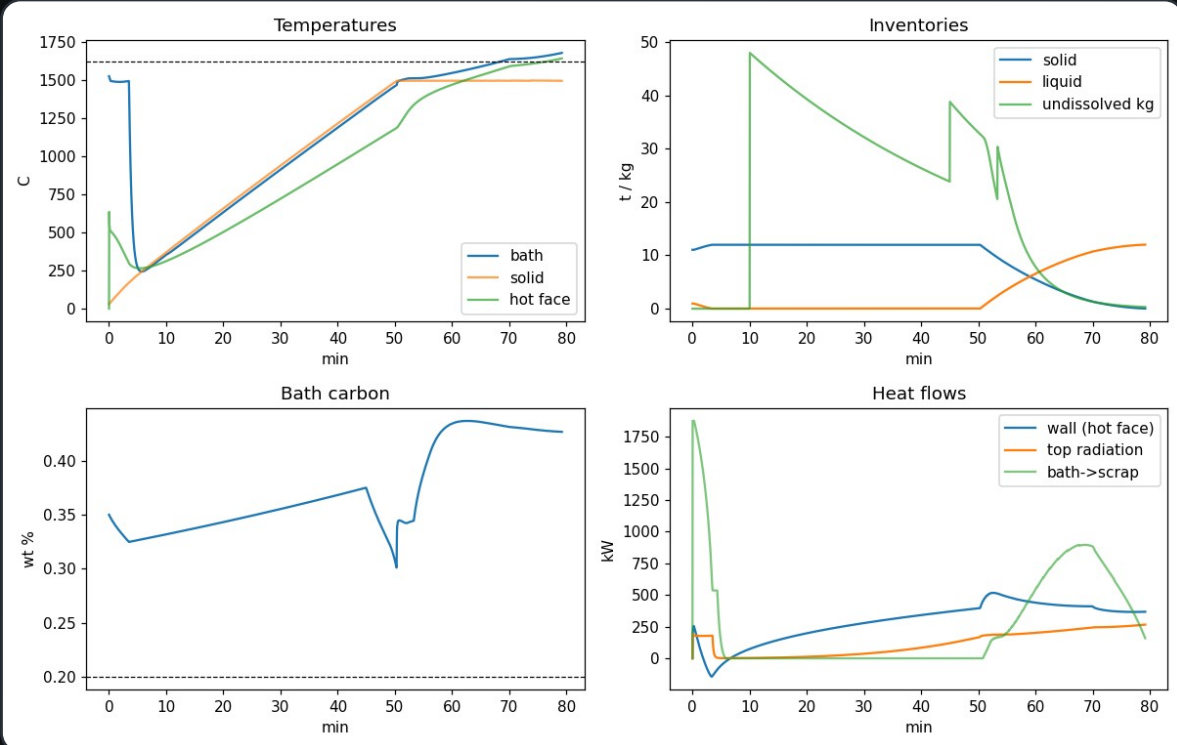
Closed-loop PLC write path — only after the advisory log demonstrates sustained trust

Process trajectory — one reference heat (Industry-X, 12 t IF)



Charge-in freeze and recovery, isothermal melt-down, superheat-gated dissolution, and — at 60 min — 150 kg of mill scale: $(\text{FeO}) + [\text{C}] \rightarrow \text{Fe} + \text{CO}$ (E27c) visibly cuts carbon, returns iron to the bath, and cools it (net endothermic, $\approx +101 \text{ kJ/mol CO}$). Element ledger $< 1 \%$, first-law closure $\approx 3 \%$.

Simulated heat & EKF tracking on a deliberately mismatched plant



EKF vs hidden plant ($\eta = 0.90$, worn lining):

σ_T collapses $35 \rightarrow 4$ °C on three immersion dips; $\hat{\eta}$ converges to 0.91 within one heat; final bath-temperature error +1.6 °C.

Left — the full reference heat:

temperatures, inventories with undissolved additions, bath carbon against aim, and the heat-flow split (wall, radiation, bath→scrap).